Research design

Biostatistics and Scientific Research

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Second semester- Third year

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Week 3

Outlines

- Experimental Study Design
- Observational Study Design
- Quasi-Experimental Study Design
- Qualitative Study Design
- Mixed-Methods Study Design

Research design strategy

- A **research design** is a strategy for answering your research question using empirical data.
- Creating a research design means making decisions about:
- Your overall research objectives and approach
- Whether you'll rely on primary research or secondary research
- Your sampling methods or criteria for selecting subjects
- Your data collection methods
- The procedures you'll follow to collect data
- Your data analysis methods

The first choice you need to make is whether you'll take a qualitative or quantitative approach

Qualitative approach	Quantitative approach
• Understand subjective experiences, beliefs, and concepts • Gain in-depth knowledge of a specific context or culture • Explore under-researched problems and generate new ideas	• Measure different types of variables and describe frequencies, averages, and correlations • Test hypotheses about relationships between variables • Test the effectiveness of a new treatment, program or product

Types of quantitative research designs

- Quantitative designs can be split into four main types.
- **Experimental** and **quasi-experimental** designs allow you: to test **cause-and-effect** relationships
- **Descriptive** and **correlational** designs allow you: to **measure variables and describe relationships between them.**

Types of qualitative research designs

Type of design	Purpose and characteristics
Case study	• Detailed study of a specific subject (e.g., a place, event, organization, etc). • Data can be collected using a variety of sources and methods. • Focuses on gaining a holistic understanding of the case.
Ethnography	• Detailed study of the culture of a specific community or group. • Data is collected by extended immersion and close observation. • Focuses on describing and interpreting beliefs, conventions, social dynamics, etc.
Grounded theory	• Aims to develop a theory inductively by systematically analyzing qualitative data.
Phenomenology	• Aims to understand a phenomenon or event by describing participants' lived experiences.

Study Design

- When choosing a study design, researchers consider **factors** such as:

1. **Research question**: What type of question am I trying to answer?

2. **Study population**: Who am I studying?

3. **Data collection methods**: What methods will I use to collect data?

4. **Resources**: What resources (time, money, personnel) do I have available?

5. **Ethical considerations**: What are the potential risks and benefits to participants?

Study Design

- **Study design** refers to a research study's overall plan and structure, outlining how the research question will be addressed and the methods used to collect and analyze the data.

- **Common types of study designs:**

1. Experimental Study Design

2. Observational Study Design

3. Quasi-Experimental Study Design

4. Qualitative Study Design

5. Mixed-Methods Study Design

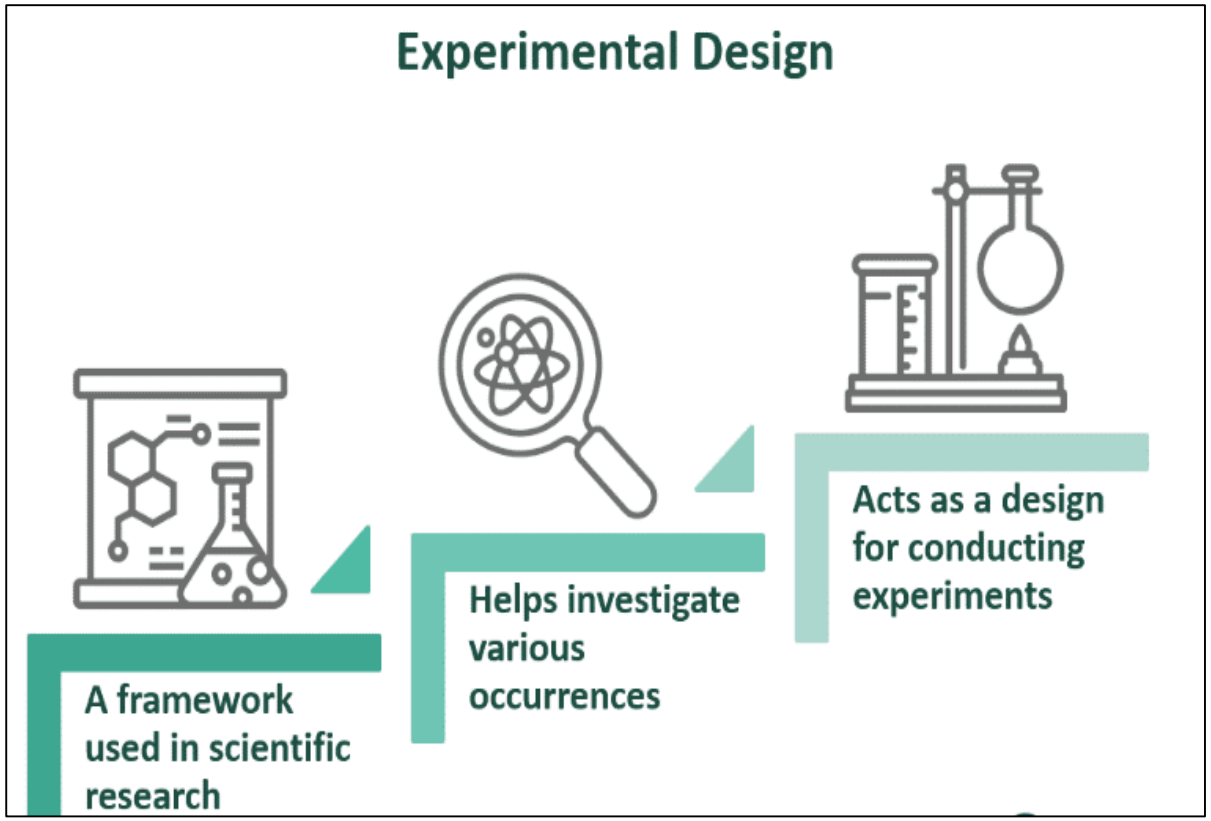
1. Experimental Study Design

❑ **Randomized Controlled Trial (RCT):**

Participants are randomly assigned to either an intervention or control group.

❑**Non-Randomized Controlled Trial:**

Participants are assigned to groups based on non-random criteria.



Experimental Study in Animals

Experimental Study Design Components for Animals

1. Treatment Groups: Animals are divided into groups based on the treatment they receive.

2. Control Groups: Animals that do not receive the treatment are used as a comparison to the treatment groups.

3. Randomization: Animals are randomly assigned to treatment or control groups to minimize bias.

4. Blinding: Researchers may be blinded to the treatment assignments to reduce bias.

5. Sample Size: The number of animals needed to detect statistically significant differences between groups.

Experimental Study in Animals

- **Considerations for Experimental Study Design in Animals**

1. **Welfare of Animals**: Ensure the study design minimizes animal suffering and distress.

2. **Statistical Power**: Ensure the sample size is sufficient to detect statistically significant differences between groups.

3. **Control of Variables**: Ensure that all variables that could affect the outcome are controlled.

4. **Reproducibility**: Ensure that the study design allows for reproducibility of the results.

Experimental Study in Animals

- **Examples of Experimental Study Designs for Animals**

1. **Toxicity Study**: A randomized controlled trial to evaluate the toxicity of a new chemical compound in mice.

2. **Efficacy Study**: A non-randomized controlled trial to evaluate the efficacy of a new vaccine in dogs.

3. **Pharmacokinetic Study**: A crossover design study to evaluate the pharmacokinetics of a new drug in rats.

Experimental Study in Animals

- **Regulatory Guidelines for Experimental Study Design in Animals**

1. IACUC (Institutional Animal Care and Use Committee): Ensure that the study design is **approved** by the **IACUC**.

2. Good Laboratory Practice (**GLP**): Ensure the study design follows GLP **guidelines**.

3. Animal Welfare Act: Ensure that the study design complies with the Animal Welfare Act.

Experimental Study in Humans

- **Considerations for Experimental Study Design in Humans**

1. **Ethics**: Ensure the study design adheres to ethical principles, such as respect for persons, beneficence, no maleficence, and autonomy.

2. **Safety**: Ensure that the study design minimizes risks to participants.

3. **Validity**: Ensure that the study design produces valid results.

4. **Reliability**: Ensure that the study design produces reliable results.

5. **Generalizability**: Ensure that the study design allows for generalization of the results to the target population.

Experimental Study in Humans

- **Examples of Experimental Study Designs in Humans**

1. **Clinical Trial**: A randomized controlled trial to evaluate the efficacy of a new medication for treating hypertension.

2. **Behavioral Intervention Study**: A non-randomized controlled trial to evaluate the effectiveness of a new behavioral intervention for weight loss.

3. **Nutrition Study**: A crossover design study to evaluate the effects of a new dietary supplement on cardiovascular health.

Experimental Study in Humans

- **Regulatory Guidelines for Experimental Study Design in Humans**

1. **Institutional Review Board** (IRB): Ensure that the study design is approved by the IRB.

2. **Good Clinical Practice** (GCP): Ensure the study design follows GCP guidelines.

3. **Helsinki Declaration**: Ensure that the study design adheres to the principles of the Helsinki Declaration.

Declaration of Helsinki

- **Guiding Principle:**

1. Protecting Patient Health.
2. Knowledge Cannot Trample Rights.
3. Additional Considerations.
4. Following Local Regulatory Norms.
5. Consideration of Risks, Burdens, and Benefits.
6. Protection of Vulnerable Groups and Individuals.
7. Focus on Ethics.
8. Protection of Privacy and Confidentiality.

2- Observational Study Design

- **An observational study** examines the effect of some intervention, risk, diagnostic test or treatment without trying to manipulate who is exposed to it or isn't.
- **Cohort Study**: A group of people with a shared characteristic are followed over time to observe outcomes.
- **Case-Control Study**: People with a specific outcome are compared to those without the outcome to identify potential risk factors.
- **Cross-Sectional Study**: A snapshot of a population at a single time to examine relationships between variables.

3- Quasi-Experimental Study Design

- Before-After Study: Measures are taken before and after an intervention to assess its impact.
- Time-Series Study: Measures are taken at multiple times to examine trends and patterns.

4- Qualitative Study Design

- **Phenomenological Study**: In-depth interviews or observations are used to explore people's experiences and perceptions.
- **Grounded Theory Study**: Data are collected and analyzed to develop a theoretical framework that explains a phenomenon.
- **Content Analysis**: Text or image data are analyzed to identify themes, patterns, and meanings.

5- Mixed-Methods Study Design

- Combination of Quantitative and Qualitative
- **Methods**: Both numerical data and textual or observational data are collected and analyzed to provide a more comprehensive understanding of the research question.

THANK
YOU